

Software Engineering – UE24CS341A

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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Section : F

Project: OpenAPI Specification & Mock Generation :

Epic 1: OpenAPI Specification & Mock Generation

Description:

Import, parse, and process OpenAPI specifications to generate schema-compliant mock responses and support API mocking.

User Stories:

- LP-5 – Import OpenAPI 3.0 Specification — **5 points**
- LP-6 – Parse OpenAPI Specification — **8 points**
- LP-7 – Generate Schema-Compliant Mock Responses — **8 points**

Total Story Points: 21

Epic 2: Request Validation & Mock Responses

Description:

Validate incoming API requests against the defined schema and return appropriate mock responses.

User Stories:

- LP-8 – Handle Mock API Requests — **5 points**
- LP-9 – Validate API Requests Against Schema — **8 points**
- LP-10 – Return Appropriate Mock Responses — **5 points**

Total Story Points: 18

Epic 3: Endpoint Configuration & Visibility

Description:

Configure simulated API endpoint behaviour and provide visibility into generated mock endpoints.

User Stories:

- LP-11 – Configure Simulated Response Latency — **3 points**
- LP-12 – View Generated Mock Endpoints — **2 points**

Total Story Points: 5

Epic 4: Performance & Reliability

Description:

Ensure that generated mock endpoints remain available and can handle high-volume API requests reliably.

User Stories:

- LP-13 – Handle High-Volume API Requests — **13 points**
- LP-14 – Maintain Mock Endpoint Availability — **5 points**

Total Story Points: 18

Sprint Execution

Briefly document what you did:

A one-week sprint was created in Jira and selected user stories were added to the sprint. The sprint was started and the tasks were progressed through the Agile workflow from **To Do → In Progress → Done**. The sprint was used to simulate the completion of planned work and observe team progress.

The PDF explicitly instructs students to create a one-week Sprint 1, add selected stories, start the sprint, and move tasks through **To Do → In Progress → Done**.

Story Point Estimation

The lab requires assigning story points using the **Fibonacci sequence**. The PDF specifically gives the sequence as 0, 1, 2, 3, 5, 8, 13, 20, 40, 100... and explains that it represents increasing uncertainty and complexity.

Story Story Points

LP-5 5

LP-6 8

LP-7 8

LP-8 5

LP-9 8

LP-10 5

LP-11 3

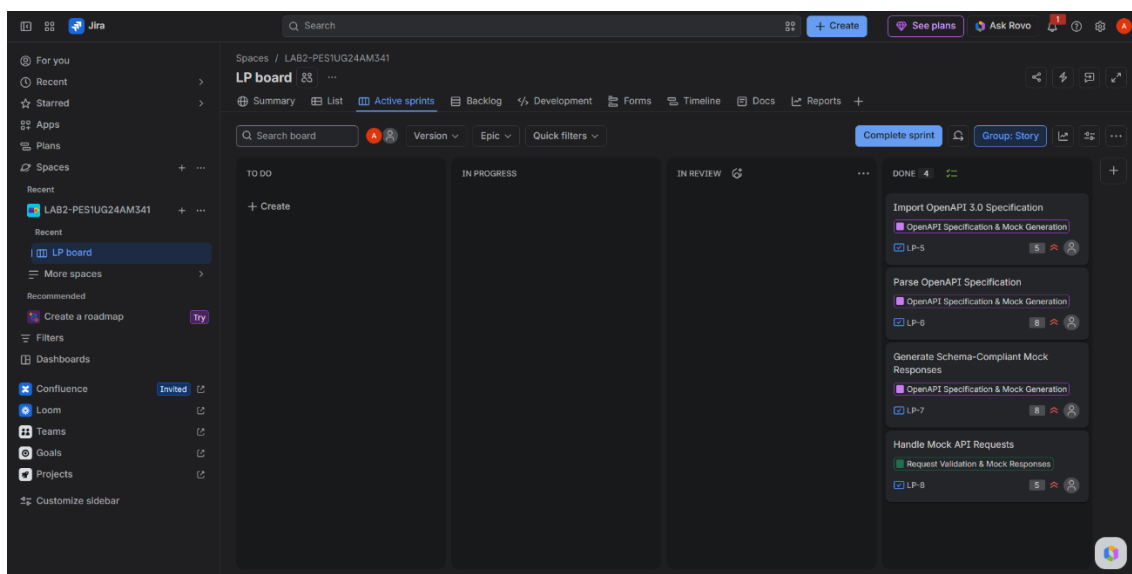
LP-12 2

LP-13 13

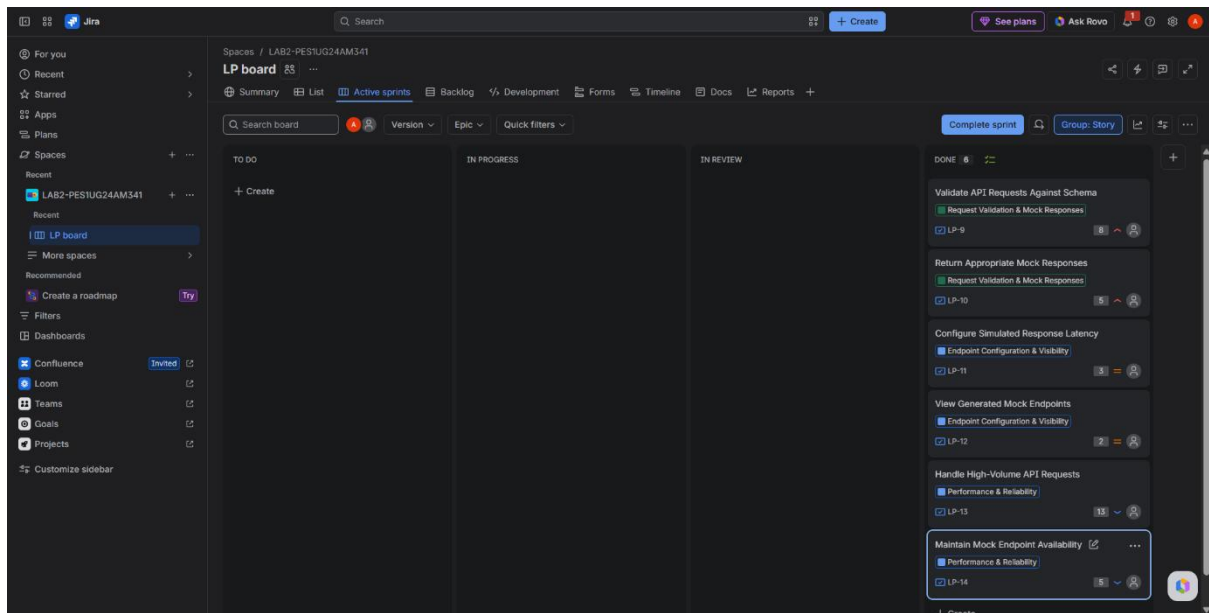
LP-14 5

Total = 62 Story Points.

JIRA Sprint 1 Completion Screenshot :



JIRA Sprint 2 Completion Screenshot :

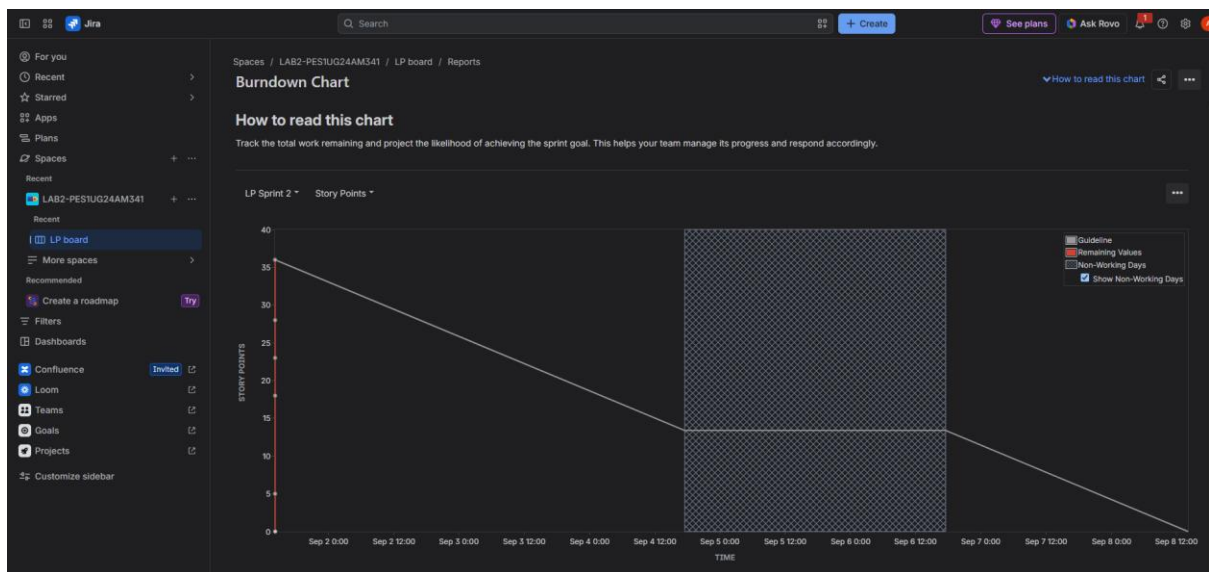


JIRA All Epics and Stories with Story Points :

This screenshot shows the Jira LP board for the space LAB2-PESTUG24AM341, displaying a list of all epics and stories with their story points. The board is titled 'LP board' and has tabs for Summary, List, Active sprints, Backlog, Development, Forms, Timeline, Docs, Reports, and a plus icon. The 'List' tab is selected. The board is divided into four columns: TO DO, IN PROGRESS, IN REVIEW, and DONE. The 'TO DO' column contains a list of items, each with a title, a status bar, and a link to the item. The items are: 'OpenAPI Specification & Mock Generation' (LP-1, Medium, To Do, 5), 'Import OpenAPI 3.0 Specification' (LP-2, Highest, Done, 5), 'Parse OpenAPI Specification' (LP-3, Highest, Done, 5), 'Generate Schema-Compliant Mock Responses' (LP-4, Highest, Done, 5), 'Request Validation & Mock Responses' (LP-5, Medium, To Do, 5), 'Handle Mock API Requests' (LP-6, Highest, Done, 5), 'Validate API Requests Against Schema' (LP-7, High, Done, 5), 'Return Appropriate Mock Responses' (LP-8, High, Done, 5), 'Endpoint Configuration & Visibility' (LP-9, Medium, To Do, 5), 'Configure Simulated Response Latency' (LP-10, Medium, Done, 5), 'View Generated Mock Endpoints' (LP-11, Medium, Done, 2), 'Performance & Reliability' (LP-12, Medium, To Do, 5), 'Handle High-Volume API Requests' (LP-13, Low, Done, 15), and 'Maintain Mock Endpoint Availability' (LP-14, Low, Done, 5). The 'Create' button is visible at the bottom of the 'TO DO' column.

Work	Assignee	Reporter	Priority	Status	Resolution	Created	Story Points
OpenAPI Specification & Mock Generation	Unassigned	Anand	Medium	To Do	Unresolved	Sep 01, 2026, 11:07 AM	None
Import OpenAPI 3.0 Specification	Unassigned	Anand	Highest	Done	Unresolved	Sep 01, 2026, 11:15 AM	5
Parse OpenAPI Specification	Unassigned	Anand	Highest	Done	Unresolved	Sep 01, 2026, 11:17 AM	5
Generate Schema-Compliant Mock Responses	Unassigned	Anand	Highest	Done	Unresolved	Sep 01, 2026, 11:25 AM	5
Request Validation & Mock Responses	Unassigned	Anand	Medium	To Do	Unresolved	Sep 01, 2026, 11:08 AM	None
Handle Mock API Requests	Unassigned	Anand	Highest	Done	Unresolved	Sep 01, 2026, 11:27 AM	5
Validate API Requests Against Schema	Unassigned	Anand	High	Done	Unresolved	Sep 01, 2026, 11:27 AM	5
Return Appropriate Mock Responses	Unassigned	Anand	High	Done	Unresolved	Sep 01, 2026, 11:28 AM	5
Endpoint Configuration & Visibility	Unassigned	Anand	Medium	To Do	Unresolved	Sep 01, 2026, 11:09 AM	None
Configure Simulated Response Latency	Unassigned	Anand	Medium	Done	Unresolved	Sep 01, 2026, 11:30 AM	5
View Generated Mock Endpoints	Unassigned	Anand	Medium	Done	Unresolved	Sep 01, 2026, 11:30 AM	2
Performance & Reliability	Unassigned	Anand	Medium	To Do	Unresolved	Sep 01, 2026, 11:11 AM	None
Handle High-Volume API Requests	Unassigned	Anand	Low	Done	Unresolved	Sep 01, 2026, 11:32 AM	15
Maintain Mock Endpoint Availability	Unassigned	Anand	Low	Done	Unresolved	Sep 01, 2026, 11:33 AM	5

Burndown Chart :



Reflection Questions :

1. Did your estimations reflect the actual effort?

Ans : Yes, the story point estimations provided a reasonable representation of the effort and complexity involved in completing the user stories. Smaller tasks such as configuring simulated response latency were assigned fewer points, while complex tasks such as handling high-volume API requests were assigned higher points. The Fibonacci scale also helped represent the uncertainty associated with more complex tasks.

2. Was your backlog well-prioritized?

Ans : Yes, the backlog was prioritized according to the importance and complexity of the functionality. Core activities such as importing and parsing the OpenAPI specification and generating mock responses were given higher priority because they form the foundation of the system. Supporting functionality such as endpoint visibility and performance improvements was handled subsequently.

3. How did your simulated sprint align with your plan?

Ans : The simulated sprint generally aligned with the planned work. The selected user stories were added to the sprint and progressed through the workflow. Most of the planned stories were completed during the simulation, demonstrating how sprint planning can be used to organize and track development work.

4. What insights did the burndown chart give about your team's capacity?

Ans : The burndown chart provided a visual representation of the remaining story points throughout the sprint. It helped compare the team's actual progress with the expected guideline. The chart showed how completed work reduced the remaining workload and provided an indication of whether the planned work could be completed within the sprint duration. It also demonstrated the importance of realistic estimation and proper sprint planning.